

Example-1: Walking at $\frac{3}{4}^{\text{th}}$ of his usual speed, a man was 25 minutes late to reach his office. Find how much time he usually takes to reach his office.

Example-2: If I walk at a speed of 3Km/hr I will reach my office 40 minutes late. However, if I walk at a speed of 4 Km/hr I will reach office 30 minutes early. Find the distance between my house and my office.

Example-3: Two friends Ram and Shyam started simultaneously from A towards B at speeds of 3kmph and 4kmph respectively. Shyam reached B first, returned immediately and met Ram at X. If the distance between A and B is 21Km, find the distance between A and X.

Example-4: One train started from Delhi at 9:00 AM at a speed of 25 Kmph. Another train started from Delhi on the same day at 1:00 PM and travelled in the same direction at 35 Kmph. At what time and how far from Delhi will the two trains meet?

Example-5: One train started from A towards B and simultaneously another train started from B towards A at speeds of 20kmph and 30 kmph respectively. After how much time and how far from B will the two trains meet? (The distance between A and B is 60 Km.)

Example-6: Two trains having lengths of 108m and 112m took 6 seconds to cross each other while travelling in opposite directions. If one of the trains was travelling at a speed of 50 kmph then find the speed of the other train in kmph.

Example-7: A train takes 20 seconds to cross a cyclist travelling in the same direction at a speed of 11 km/hr but only 9 seconds to cross a lamppost. Find the speed and the length of the train.

Example-8: A man can row with the stream at 12 km/hr and against the stream 8 km/hr. Find the rate in still water.

- (a) 10km/hr (b) 2km/hr (c) 8km/hr (d) 5km/hr

Example-9: A boat can move at a speed of 6 km/hr in still water. However, it takes twice as much time to travel a certain distance upstream as compared to the same distance downstream. Find the speed of the stream.

Example-10: A man rows upstream 16kms and downstream 31kms taking 5 hours each time. What is the velocity of current?

- (a) 5km/hr (b) $5/2$ km/hr (c) 3km/hr (d) $3/2$ km/hr

Example-11: The current of stream runs at the rate of 5km/hr. A motor boat goes 10km up stream and back again to its starting point in 50 min. Find the speed of motor boat in still water.

- (a) 30km/hr (b) 25km/hr (c) 20km/hr (d) None

Example-12: A can beat B by 20 meters and B can beat C by 20 meters in a race of 100 meters. A will beat C by how many meters in a race of

- (a) 100 m
(b) 150 m

Example-13: In a 500m race, A beats B by 20m or 5 seconds. What is the speed of A?

- (a) 4m/sec (b) $25/3$ m/sec (c) $25/6$ m/sec (d) 5 m/sec